DELHI PUBLIC SCHOOL JANKIPURAM
ANNUAL EXAMINATION: (2024-2025)
SUBJECT: SCIENCE (086)
GRADE: IX

SET–I

Time – 3 hrs. **M.M. 80**

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General Instructions:

Read the following instructions carefully and follow them:

- This question paper consists of 39 questions in 5 sections
- All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.
- Section A consists of 20 objective type questions carrying 1 mark each
- Section B consists of 6 Very Short questions carrying 02 marks each.
- Section C consists of 7 Short Answer type questions carrying 03 marks each.
- Section D consists of 3 Long Answer type questions carrying 05 marks each.
- Section E consists of 3 source-based/case-based units of assessment of 04 marks each with sub-parts.

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SECTION – A
OBJECTIVE BASED QUESTIONS (1×20=20)

Choose the correct option:

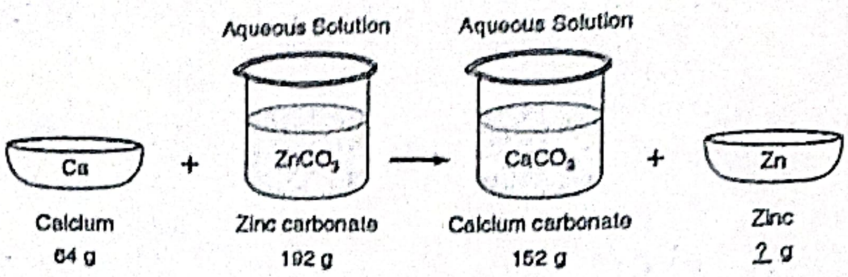
1. In which of the following set up, the rate of diffusion is faster?
   a) 5g of Potassium permanganate crystals in 100ml of water at 25 °C.
   b) 5g of Potassium permanganate crystals in 100ml of water at 0 °C.
   c) 5g of Potassium permanganate crystals in 100ml of water at 15 °C.
   d) 5g of Potassium permanganate crystals in 100ml of water at 60 °C.

2. A solution in which no more solute can be dissolved at a given temperature is known as
   a) True solution
   b) Unsaturated solution
   c) Saturated solution
   d) Dilute Solution

3. The weight of an object at the centre of the earth of radius R is
   a) Zero
   b) Infinite
   c) R times the weight at the surface of the Earth
   d) $1/R^2$ times the weight at the surface of the Earth

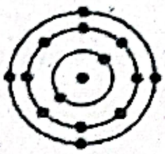
4. According to law of conservation of mass, how much Zinc was present in the Zinc carbonate

a) 40g
b) 88g
c) 104g
d) 256g



5. Schematic representation of an element is shown, Identify the element:

a) Na
b) S
c) Si
d) Sn.



6. How does Endoplasmic reticulum helps in transporting protein between various region of the Cytoplasm:

a) by forming a network of membrane bound tubes in the cytoplasm
b) by occupying most of the space in the cytoplasm
c) by generating small transport vesicles throughout the cell
d) by directing all cell organelles to perform the same biochemical activity.

7. In summer, the leaves of a potted plant droop when the soil becomes dry. Which cell organelle makes the leaves droop?

a) Nucleus
b) Cell wall
c) Lysosomes
d) Vacuole

8. When a body falls freely towards the earth, then its total energy

a) increases
b) decreases
c) remains constant
d) first increases and then decreases

9. While doing work and running, you move your organs like hands, legs etc. Which among the following is correct?

a) Smooth muscles contract and pull the ligament to move the bones
b) Smooth muscles contract and pull the tendons to move the bones
c) Skeletal muscles contract and pull the ligament to move the bones
d) Skeletal muscles contract and pull the tendon to move the bones

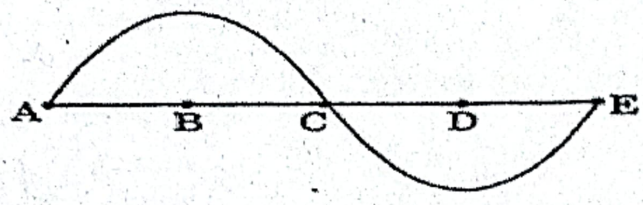
10. Which of the following cells is found in the cartilaginous tissue of the body?

a) Mast cells.
b) Basophils
c) Osteocytes.
d) Chondrocytes

11. Momentum of a body increases from 20kgms^{-1} to 40kgms^{-1} in 5 seconds, then the force applied is
    a) -4 N.
    b) 4 N.
    c) 20 N.
    d) -20 N

12. Which of the following can sometimes be Zero for a moving body:
    i) Distance travelled ii) Average velocity
    iii) Average speed iv) Displacement
    a) Only (i)
    b) (i) and (ii)
    c) (ii) and (iv)
    d) Only (iv)

13. In the curve half the wavelength is :



    a) AB
    b) BD
    c) CD
    d) AE

14. If a car is traveling at 30 m/s and comes to a stop in 10 seconds, what is its acceleration?
    a) -3 m/s^2
    b) -10 m/s^2
    c) -5 m/s^2
    d) -2 m/s^2

15. Observe the figure A and B and choose the correct option

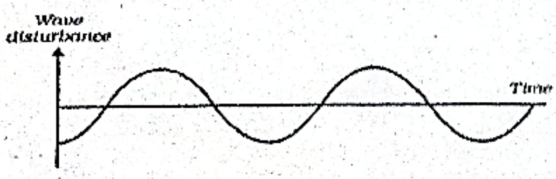


Fig.A

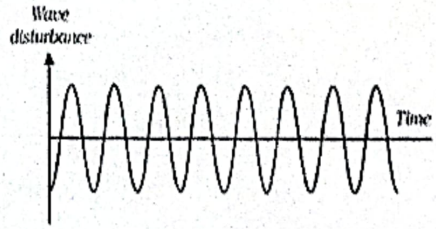


Fig. B

    a) A represents low pitch, high frequency
    b) A represents high pitch, low frequency
    c) B represents high pitch, low frequency
    d) B represents low pitch, low frequency

16. A sound wave with a frequency of 2000 Hz and a wavelength of 0.35 meters, what is the speed of the sound wave?

a. 70 m/s
b. 700 m/s
c. 7 m/s
d. 0.7 m/s

Q. No. 17 to 20 are Assertion – Reasoning based questions:-
These consists of two statements – Assertion (A) and Reason (R). Answer the questions selecting the appropriate option given below :

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true and R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

17. Assertion: A sodium ion has positive charge.
Reason: Sodium ion has more protons than a neutral atom.

18. Assertion (A): Water hyacinth can float on water surface.
Reason (R): Aerenchyma tissue is present in water hyacinth.

19. Assertion: Universal Gravitational constant G is a scalar quantity
Reason : The value of G is same throughout the Universe

20. Assertion : While walking on ice, one should take small steps to avoid slipping.
Reason : Smaller steps ensures less friction.

SECTION – B
VERY SHORT ANSWER TYPE QUESTIONS (2×6=12)

21. Show the interconversion of three states of matter with the help of flow chart.

22. How does fungi and bacteria withstand much greater changes in the surrounding medium than animal cells?

23. i) How are messages conveyed from one place to another within the body?
ii) How are glandular epithelium formed?

OR

a) Name the tissue present in:
i) lining of blood vessels
ii) lining of small intestine

b) Name the fat storing tissue in the animals and mention its function.

24. Give reasons:
i) We get a crunchy and granular feeling, when we chew guava fruit.
ii) Branches of a tree move and bend freely in high wind velocity

OR

a) What is the specific function of the cardiac muscle?
b) Name the regions in which parenchyma tissue is present.

25. Water sprinkler used for grass lawns begins to rotate as soon as the water is supplied. Which law of motion is the cause of rotation of the water sprinkler? Explain.

26. A tomato plant in a farmer's field shows signs of stunted growth and yellowing of leaves. Upon soil testing, it was found that nitrogen and magnesium were deficient.
(a) Classify the nutrients nitrogen and magnesium as macronutrient or micronutrient.
(b) Suggest a remedy to replenish these nutrients in the soil.

OR

Give two set of contrasting features between Apiculture and Aquaculture.

SECTION – C
SHORT ANSWER TYPE QUESTIONS (3×7=21)

27. Non metals are usually poor conductors of heat and electricity. They are non-lustrous, non-sonorous, non-malleable and are coloured.

(a) Name a lustrous non-metal.

(b) Name a non-metal which exists as a liquid at room temperature.

(c) The allotropic form of a non-metal is a good conductor of electricity. Name the allotrope.

(d) Name a non-metal which is known to form the largest number of compounds.

(e) Name a non-metal other than carbon which shows allotropy.

(f) Name a non-metal which is required for combustion.

28. Write the formula and name of the compound formed by combination of

(atomic mass of Fe=56; S=32; O=16; N=14; C=12; H=1)

i) Fe^{3+} and SO_4^{2-}

ii) NH_4^+ and CO_3^{2-}

Also calculate their molecular mass.

29. An atom of an element has two electrons in outermost M-shell. State its
i) Name
ii) electronic configuration
iii) Valency and make its formula with sulphide ion.

OR

Complete the following table:

Atom	Mass no.	Atomic no.	No. of neutrons
$^{60}Co_{27}$	60	27	33
$^{23}Na_{11}$	23	11	12
$^{37}Cl_{17}$	37	17	20

30. A de-shelled egg is place in pure water and later in concentrated salt solution.
(i) Write your observation in each case and justify.
(ii) Identify and explain the process involved in the above activity.

31. (i) If a potted plant is covered with a glass jar, water vapours appear on the wall of glass jar. Identify and define the process taking place.
(ii) Draw the well labelled diagram of the part through which the above process takes place.

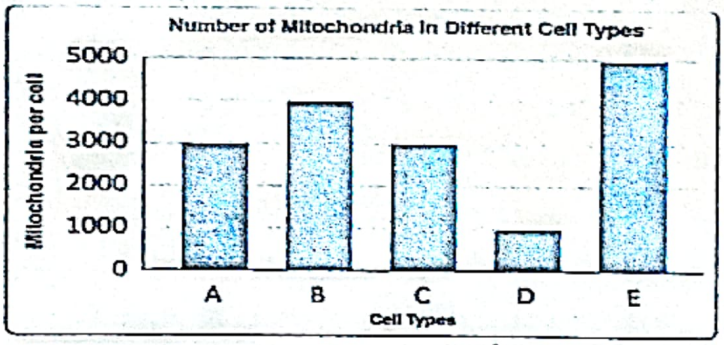
32.(a) A stone is thrown in a pond. 12 full ripples are produced in 1 second. If the distance between a crest and the adjacent trough is 10 cm, calculate the wavelength and velocity of the wave.
(b) What happens to the speed of sound on a hotter day?

33.(a) Why are ceilings of concert halls and conference halls made curved?
(b) Why are the roofs and the walls of the auditoriums covered with sound absorbent materials?
(c) State two necessary conditions for echo to be heard.

SECTION – D
LONG ANSWER TYPE QUESTIONS (5×3=15)

34. Three students A, B and C prepared mixtures using chalk powder, common salt and milk respectively in water. Whose mixture:
i) would not leave residue on filter paper after filtration? Milk water
ii) would show Tyndall effect? Common Salt
iii) would give transparent/clear solution? Salt + clear ~~and~~ chalk + transparent
iv) would settle down at the bottom when left undisturbed? Chalk
v) could be filtered by filter paper? Chalk

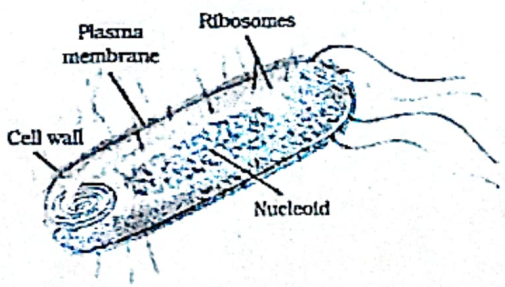
35.The graph below shows the data of five different cells and the number of mitochondria found per cell.



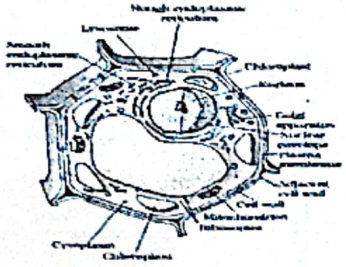
a) State the role of mitochondria.
b) Explain the structure of mitochondria.
c) What can you infer about the energy requirement of cell type E when compared to other cell types? How do you relate to the number of mitochondria? Justify your answer.

OR

Identify the type of cell shown below in fig. A and fig. B and differentiate between the two on the basis of : nuclear region , membrane bound organelles and mobility.



A



B

36. (a) Define potential energy. Derive equation for gravitational potential energy
(b) A 5kg ball is thrown upwards with a speed of 10m/s (g=10m/s).
i) Find the potential energy when it reaches the highest point.
ii) Calculate the maximum height attained by it.

OR

i) A force acting on a 10 kg mass changes its velocity from 54km/h to 90km/h. Calculate the work done by the force

ii) A body of mass 10 kg is kept at a height 10 m from the ground, when it is released after sometime its kinetic energy becomes 450 J. What will be the potential energy of the body at that instant? [g = 10m/s^2]

SECTION -E

CASE BASED AND SOURCE BASED QUESTIONS (4×3=12)

37. We define the atomic mass unit (earlier abbreviated as 'amu', but according to the latest IUPAC recommendations, it is now written as 'u' – unified mass) as equal to the mass of one carbon atom. Scientists initially took 1/16 of the mass of an atom of naturally occurring oxygen as the unit.However in 1961 for a universally accepted atomic mass unit, carbon-12 isotope was chosen as the standard reference for measuring atomic masses. One atomic mass unit is a mass unit equal to exactly one-twelfth (1/12th) the mass of one atom of carbon-12. The relative atomic masses of all elements have been found with respect to an atom of carbon-12.

a) Calculate molecular mass of HNO_3. [atomic mass: H=1; N=14; O=16]. 1
b) An element X has valency +3. Find the formula of it chloride. If the valency of X is changed to +5, then what will be the formula of its chloride. 2
c) Write the formula of sodium oxide and sodium sulphide. 1

OR

Give example of any two polyatomic ions? $SO_2 + SO_2$ $2SO$

38. While drawing water from a well, a bucket of water appears to be lighter when as it's in water due to buoyancy. The upward force exerted by the water on the bucket is known as upthrust or buoyant force. In fact, all objects experience a force of buoyancy when they are immersed in a fluid. The magnitude of this buoyant force depends on the density of the fluid.

i) Knives have sharp edges. Why? 1
ii) The volume of 50 g of a substance is 20 cm^3. If the density of water is 1 g cm^{-3}, will the substance float or sink? 2
iii) Explain the principle used in designing ships and submarine? 1

OR

Define Density. Compare the density of iron nail and cork with density of water.

39. Food security depends on both availability of food and access to it. The majority of our population depends on agriculture for their livelihood. Increasing the incomes of people working in agriculture is therefore necessary to combat the problem of hunger. Scientific management practices should be undertaken to obtain high yields from farms. For sustained livelihood one should undertake mixed farming, intercropping, and integrated farming practices, for example, combine agriculture with livestock/poultry/fisheries/bee-keeping.

i) What are fodder crops? Give two examples of such crops. 1
ii) What are the desirable agronomic characteristics for crop improvements? 2
iii) Differentiate between the kharif season and rabi season. 1

OR

What is organic farming?